420-D04-AB — Conformity Assessment

Lab 0 — Install & verify the auditing tools (tool set-up)

Cohort CYB-01 · Fall 2026 · Day 1 — Tuesday, October 27, 2026 · Professor's copy

Item	Value
Weight	**Ungraded — required.** Labs 4 and 5 cannot start without a working toolbox; the checklist is re-verified in class on Day 6 (Nov 10).
In-class time	Day 1, 19:45–22:00 (first 30–45 min guided step by step); finishing time on Day 2 (Oct 28).
Deliverable	Half-page **Environment checklist** (tool · version · VM · status · screenshot) as PDF + evidence folder (zip). File names: `LastName_FirstName_420-D04-AB_Lab0_Environment-Checklist.pdf` and `..._Lab0_Evidence.zip`.
Submission	**Omnivox LÉA → 420-D04-AB → Assignments → Lab 0 → DropBox**, before Day 7: **Tuesday, November 10, 2026, 23:59 EST** (Eastern Standard Time — clocks change on Nov 1). Earlier is better: Lab 4 (Day 7, Nov 11) uses these tools.
Where it lands	**Chapter 0 — Lab environment & tools** of your Course Procedure Manual (CPM, 10 %).
Lab base	The Day 1 real-life examples already given to you: `verify_tools.sh`, `Verify-Tools.ps1`, `environment_checklist_template.csv`, and (bonus) `Check-ScreenLock.ps1`.

0. Legal notice, disclaimer and acceptable use — read before anything runs

1. **Authorized targets only.** Every tool in this lab runs **only** on your own two lab **VMs** (virtual machines) and the fictional Érable Cycles case file — never on college, employer, home, cyber-range shared infrastructure or any third-party system.
2. **Written mandate.** You work under the instructor's written course mandate (Labs 0–2) and, from Lab 3, under the CEO authorization letter you sign. No mandate, no test.
3. **The law.** Unauthorized access to or use of a computer system is a criminal offence: Canada *Criminal Code* s. 342.1 (unauthorized use of computer) and s. 430(1.1) (mischief in relation to computer data); USA 18 U.S.C. § 1030 (Computer Fraud and Abuse Act); EU Directive 2013/40/EU; Spain *Ley Orgánica* 10/1995 (arts. 197 bis, 264). Intent to learn is not a defence.
4. **Confidentiality.** Results are confidential; screenshots must mask real personal information (names, e-mails, real IP addresses); reports go only to the instructor through LÉA.
5. **Professional ethics** (ISO 19011:2026 audit principles): an honest "not done — reason: …" is acceptable; an invented screenshot is academic misconduct (John Abbott College IPESA Policy No. 7, arts. 9.1–9.2; Policy 13).

6. **Tool hygiene.** Download only from the official links in section 7; verify the download's checksum when the vendor publishes one; keep the VMs on an isolated network (VirtualBox *Host-only* or *NAT*); never scan from the host machine.
7. **Disclaimer & IP.** Educational use within the AEC LEA.D8 program; not legal advice; no warranty. All materials © Cyber.SoHo Educational Hub (sole IP owner); personal study only. Late submissions: −10 % per calendar day (departmental policy) — not applicable to this ungraded lab but applicable to Labs 1–5.
8. **Acknowledgement.** The e-signature acknowledgement form of this package is provided separately and must be signed before Lab 4.

1. What you are doing and why

A conformity assessment compares what exists with what is required (Lesson 1). Before auditing Érable Cycles you need an **audit toolbox** that demonstrably works. Lab 0 applies the Lesson 1 chain to the toolbox itself: *requirement* (tool installed) → *test* (version command or benchmark detection) → *evidence* (screenshot, CSV row) → *finding* (FOUND / MISSING). The result is the Environment checklist — the first page of your CPM.

2. Newbie glossary (terms used below)

- **VM (virtual machine)** — a complete computer (operating system + disks) running as a program inside another computer. **Hypervisor** — the program that runs VMs (Oracle VirtualBox, VMware Workstation, the CyberQuébec cyber range).
- **ISO** — a single file containing the full installation disc of an operating system. **Evaluation edition** — a free, time-limited (90-day) licensed copy of Windows for testing.
- **Snapshot** — a saved state of a VM that can be restored in seconds; "clean" = the state before any tool was installed.
- **LTS (Long-Term Support)** — an Ubuntu release maintained for five years (24.04 = April 2024; 26.04 = April 2026).
- **Package manager (apt)** — the Ubuntu command that downloads and installs software from trusted repositories; `sudo` runs a command with administrator rights.
- **CLI / GUI** — command-line interface (you type) / graphical user interface (you click).
- **Benchmark** (CIS) — a published list of secure settings for one product (e.g. Windows 11 Enterprise). **Baseline** (Microsoft) — Microsoft's recommended security settings for a Windows version. **Hardening index** (Lynis) — a 0–100 score of how hardened a Linux system is.
- **Registry** — the Windows database of settings; **Group Policy / local policy** — the way Windows enforces settings.
- **PowerShell** — the Windows scripting shell; an **execution policy** decides whether scripts may run; `-Scope Process Bypass` allows them for the current window only.
- **Bash** — the Linux shell; `chmod +x file` makes a script executable.
- **CSV (Comma-Separated Values)** — a plain-text spreadsheet; **evidence** — a dated, named file that proves a result; **checksum (SHA-256)** — a fingerprint used to verify a download is intact and genuine.

3. Lab environment (per student)

VM	Operating system	Why	Where
VM 1	**Windows 11 Enterprise, version 26H2** (or **LTSC 2024**) — free 90-day evaluation ISO (x64 or Arm64) from the Microsoft Evaluation Center; a Microsoft account is required at first sign-in. The Windows 10 Enterprise evaluation is **no longer offered** (Windows 10 end of support: Oct 14, 2025)	CIS-CAT Lite detects the Windows **Enterprise** benchmarks (Windows 11 Enterprise Benchmark v5.1.0); Policy Analyzer compares with the Windows 11 26H2 baseline	CyberQuébec range when provisioned, otherwise VirtualBox 7.2.22 / VMware Workstation Pro 26H1 (free) on your computer (4 GB RAM, 64 GB disk, 2 vCPU)
VM 2	**Ubuntu Desktop 24.04.5 LTS** (ISO at releases.ubuntu.com/noble — matches the case company's AWS server) or **26.04.1 LTS** (the default download since April 2026)	Lynis works on every Ubuntu release; the Lite edition of CIS-CAT covers only Ubuntu 20.04	same (4 GB RAM, 30 GB disk)

Network: both VMs on the same VirtualBox **NAT Network** (or host-only network) so they can reach each other (needed from Lab 2) but are isolated from your home network. Apple-silicon Macs: use the **Arm64** Windows 11 Enterprise evaluation ISO in UTM or VMware Fusion (free), or the CyberQuébec range (ask the instructor on Day 1).

4. Step-by-step procedure (use the 7-line skeleton of the CPM for each step: purpose · prerequisites · steps · expected output · interpretation · evidence · common problems)

Step 1 — Clean snapshots (both VMs, 5 min). Purpose: a safe return point. VirtualBox: select the VM → menu button next to its name → **Snapshots → Take** → name `clean`, description "Day 1 – nothing installed yet" → OK. VMware Workstation: *VM → Snapshot → Take Snapshot*. CyberQuébec: *VM page → Snapshots*. Expected: a snapshot named `clean` marked *Current State*. Evidence: `L0-E01_snapshot-win.png`, `L0-E02_snapshot-ubuntu.png`. Problem: VM must be powered off or paused on some hypervisors — power off first.

Step 2 — Ubuntu VM: install Lynis and Nmap (10 min). Terminal (`Ctrl+Alt+T`):

```
sudo apt update && sudo apt install -y lynis nmap     # install both tools from the Ubuntu repositories
lynis --version                                       # expected: 3.0.9 on Ubuntu 24.04 (3.1.6 on 26.04) — old but acceptable
nmap --version | head -n 1                            # expected: "Nmap version 7.94 ..." on 24.04 (7.98 on 26.04)
```

Recommended — the current Lynis **3.1.7** (June 25, 2026) instead of the old apt package: either `git clone https://github.com/CISOfy/lynis && cd lynis && sudo ./lynis audit`

`system`, or add the CISOfy repository (`curl -fsSL https://packages.cisofy.com/keys/cisofy-software-public.key | sudo gpg --dearmor -o /etc/apt/trusted.gpg.d/cisofy-software-public.gpg`, then `echo "deb [arch=amd64,arm64 signed-by=/etc/apt/trusted.gpg.d/cisofy-software-public.gpg] https://packages.cisofy.com/community/lynis/deb/ stable main" | sudo tee /etc/apt/sources.list.d/cisofy-lynis.list`, then `sudo apt update && sudo apt install lynis`). Lynis warns "release is old" with the 3.0.9 package — write the version you actually have. Evidence: `L0-E10_os-release.png` (`cat /etc/os-release | head -n 2`), `L0-E11_lynis-version.png`, `L0-E12_nmap-version.png`. Problem: "Unable to locate package" → run `sudo apt update` again or check the VM has Internet (NAT).

Step 3 — Ubuntu VM: first Lynis audit (5 min, read-only).

```
mkdir -p ~/lab0                                          # folder for this lab's evidence
sudo lynis audit system                                  # ~300 tests, 1–3 minutes, changes nothing
sudo cp /var/log/lynis-report.dat ~/lab0/L0-E14_lynis-report.dat   # keep the report as evidence
```

Expected: the last block shows `Hardening index : NN [####...]`, the number of warnings and suggestions, and the report path. Write the index in the checklist; **no analysis tonight** (Lab 4). Evidence: `L0-E13_lynis-index.png`. Problem: "permission denied" → you forgot `sudo`.

Step 4 — Windows VM: CIS-CAT Lite (20 min). 1) Register (free) at the CIS-CAT Lite page (link in section 7), download `CIS-CAT Lite` (it is CIS-CAT Pro Assessor **v4.67.0**, October 5, 2026, running without a licence; Java 21 is bundled in the Windows build), 2) unzip into `C:\Tools\` (right-click → *Extract All*; never run from inside the zip), 3) open an **Administrator** Command Prompt, go to the unzipped folder and start `Assessor-GUI.bat` (or `Assessor-CLI.bat -i` for the interactive text version), 4) confirm the benchmark **CIS Microsoft Windows 11 Enterprise Benchmark** (or Windows 10 Enterprise) is detected, choose the **Level 1** profile, enable the **HTML** report, run once (≈ 5 min), 5) note the score and save the report. Expected: a report in the `reports` sub-folder with a pass/fail per recommendation and a score. Evidence: `L0-E23_ciscat.png` (benchmark list), `L0-E25_ciscat_report.html`. Problems: Java error → the Windows download bundles Java 21, re-extract the whole zip; benchmark not listed or "benchmark autodetect error" on Windows 11 → choose the Windows 11 Enterprise benchmark manually in the list and check `winver` shows *Enterprise*; Defender SmartScreen warning → *More info* → *Run anyway* (the file came from cisecurity.org). Note: the Lite edition does **not** cover Ubuntu 22.04/24.04/26.04 (only 20.04) — that is why Linux uses Lynis.

Step 5 — Windows VM: Security Compliance Toolkit & Policy Analyzer (15 min). 1) On the Microsoft Download Center page of the **Security Compliance Toolkit 1.0** (section 7) select `PolicyAnalyzer.zip` and the **Windows 11 version 26H2 Security Baseline v2** zip (toolkit page updated October 8, 2026; 25H2 / 24H2 baselines are also listed — take the one matching your VM, `winver` shows the version), 2) unzip both into `C:\Tools\SCT\`, 3) start `PolicyAnalyzer\PolicyAnalyzer.exe` (portable, no installer; Policy Analyzer 4.0.x), 4) **Add…** → *File* → *Add files from GPO(s)…* → select the baseline's `GPOs` folder → Import →

save the `.PolicyRules` file, 5) tick the rule set → **Compare to Local Policy** → the Policy Viewer lists every setting where the VM differs from the baseline (yellow). Note three differences. Evidence: `L0-E24_policyanalyzer.png`. Problems: the baseline for your exact Windows version is not listed → take the most recent Windows 11 baseline; Policy Analyzer needs .NET Framework 4.x, present by default on Windows 11.

Step 6 — Windows VM: PowerShell checks (5 min). Start **Windows PowerShell as Administrator** (built-in 5.1; PowerShell 7.6 `pwsh` — install with `winget install --id Microsoft.PowerShell --source winget` — is optional and gives the same output):

```
Get-ComputerInfo | Select-Object OsName, OsVersion, OsBuildNumber   # slow
the first time (10-30 s) — normal
Get-HotFix | Select-Object -First 5                                  #
installed updates (Lab 4 patch check)
$PSVersionTable.PSVersion                                            #
PowerShell version
```

Evidence: `L0-E20_computerinfo.png`, `L0-E21_psversion.png`, `L0-E22_hotfix.png`.

Step 7 — Run the two verification scripts (10 min). Copy the scripts from the Day 1 downloadables.

```
chmod +x verify_tools.sh && ./verify_tools.sh          # Ubuntu VM →
~/lab0/environment_checklist_linux.csv
Set-ExecutionPolicy -Scope Process Bypass             # allow scripts in this
window only
.\Verify-Tools.ps1                                     # Windows VM →
~\lab0\environment_checklist_windows.csv
```

Expected: one row per tool with FOUND / MISSING. A **MISSING** row is a finding, not a failure: fix the cause (most often an archive that was never extracted) and re-run until all rows are FOUND. Bonus (+ a first real finding): run `Check-ScreenLock.ps1` from Lesson 1 and keep its `evidence\findings.csv`.

Step 8 — Build the Environment checklist (10 min). Open `environment_checklist_template.csv` (or the Day 1 XLSX, sheet *Environment checklist*), paste the rows from both CSV files, add the two snapshot rows and the Lynis hardening index, keep the column order *tool · version · VM · status · evidence*, export as PDF with the file name above. Put all screenshots in a folder `lab0-evidence` using exactly the names in the evidence column; mask personal information; zip the folder.

Step 9 — Submit (2 min). Omnivox → LÉA → 420-D04-AB → Assignments → **Lab 0 — Environment checklist** → DropBox → attach the PDF and the zip → Submit → screenshot the confirmation (`L0-E30_omnivox-submission.png`).

5. Self-check before submitting (what the instructor verifies on Day 6)

- [ ] Two snapshots named `clean` (screenshots). - [ ] Lynis and Nmap versions recorded; hardening index noted; report copied.
- [ ] CIS-CAT Lite detects the Windows Enterprise benchmark; one HTML report saved. - [ ] Policy Analyzer compared the Windows 11 baseline; 3 differences noted.
- [ ] `Get-ComputerInfo`, `Get-HotFix`, `$PSVersionTable` screenshots. - [ ] Both verification CSV files, all rows FOUND.
- [ ] Checklist PDF named correctly, evidence zip, personal information masked, submitted on LÉA with confirmation screenshot.

6. Alternatives (free / open source) if a tool cannot be used

Need	Course tool	Free alternatives (official sites in section 7)
Linux configuration audit	Lynis	**OpenSCAP** (`sudo apt install openscap-scanner openscap-utils ssg-base ssg-debderived` → `oscap xccdf eval --profile <id> /usr/share/xml/scap/ssg/content/ssg-ubuntu2404-ds.xml`; the Ubuntu package may ship older content — check the folder, or download the latest ComplianceAsCode release), **Ubuntu Security Guide** `usg` with CIS profiles (free personal Ubuntu Pro, 5 machines), **Wazuh** agent SCA module
Windows configuration audit	CIS-CAT Lite, Policy Analyzer	**HardeningKitty** (PowerShell; audits against CIS, Microsoft baselines, BSI lists), **LGPO** from the same Security Compliance Toolkit (apply/export local policy), **Microsoft Defender Vulnerability Management** (cloud, trial)
Vulnerability scan (Lab 4 context)	—	**Nessus Essentials** (free, now limited to 5 IPs, 30-day licence; Essentials Plus is free for one year for verified students and instructors), **Greenbone OpenVAS Community Edition**
Inventory / facts (Lab 2)	PowerShell, Bash, Nmap	**osquery** (SQL over the OS), **GLPI** / **Snipe-IT** (asset management)
Policy-as-code checks	check scripts	**CINC Auditor** (open-source build of Chef InSpec), **Prowler** and **ScoutSuite** for AWS/Azure (demo only)
Hypervisor	VirtualBox 7.2.22	**VMware Workstation Pro 26H1** (free for personal, educational and commercial use since Nov 2024 — Broadcom portal, free account), **VMware Fusion** / **UTM** (macOS, Arm64 ISO), **Hyper-V** (Windows Pro/Enterprise hosts), the **CyberQuébec** range

7. Official links (first-party domains only; re-check any link with the VirusTotal URL scanner)

- Windows 11 Enterprise evaluation (26H2 / LTSC 2024): https://www.microsoft.com/en-us/evalcenter/evaluate-windows-11-enterprise · Ubuntu 24.04.5 LTS ISO: https://releases.ubuntu.com/noble/ · Ubuntu Desktop (26.04.1 LTS): https://ubuntu.com/download/desktop · VirtualBox: https://www.virtualbox.org/wiki/Downloads · VMware Workstation Pro (Broadcom, free):

https://support.broadcom.com/group/ecx/productdownloads?subfamily=VMware%20Workstation%20Pro&freeDownloads=true

- CIS-CAT Lite (registration): https://learn.cisecurity.org/cis-cat-lite · Assessor docs (coverage, Java 21): https://ciscat-assessor.docs.cisecurity.org/en/latest/ · CIS Benchmarks: https://www.cisecurity.org/cis-benchmarks
- Microsoft Security Compliance Toolkit 1.0: https://www.microsoft.com/en-us/download/details.aspx?id=55319 · PowerShell 7 install: https://learn.microsoft.com/en-us/powershell/scripting/install/installing-powershell-on-windows
- Lynis: https://cisofy.com/lynis/ · GitHub: https://github.com/CISOfy/lynis · CISOfy apt repository: https://packages.cisofy.com/community · Nmap: https://nmap.org/download.html
- OpenSCAP: https://www.open-scap.org/ · Ubuntu Security Guide (CIS profiles): https://ubuntu.com/security/certifications/docs/usg/cis · Wazuh: https://wazuh.com/ · HardeningKitty: https://github.com/scipag/HardeningKitty · Nessus Essentials: https://www.tenable.com/products/nessus/nessus-essentials · Greenbone Community docs: https://greenbone.github.io/docs/latest/ · osquery: https://github.com/osquery/osquery · CINC Auditor: https://cinc.sh/docs/auditor/ · Prowler: https://github.com/prowler-cloud/prowler · ScoutSuite: https://github.com/nccgroup/ScoutSuite · Defender Vulnerability Management: https://learn.microsoft.com/en-us/defender-vulnerability-management/defender-vulnerability-management
- Omnivox (LÉA): https://johnabbott.omnivox.ca/ · CyberQuébec: https://cyberquebec.org/ · VirusTotal URL scanner: https://www.virustotal.com/gui/home/url · Criminal Code s. 342.1: https://laws-lois.justice.gc.ca/eng/acts/C-46/section-342.1.html · ISO 19011:2026 (audit principles): https://www.iso.org/standard/19011

Versions and download pages verified on October 27, 2026 — see the footnotes in the visual walkthrough if a screen looks different. Companion files: Lab 0 quick guide (2 pages) and the Lab 0 visual walkthrough (HTML with animations).

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